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Description

This project puts to work all the basics concepts of app development from frontend to backend database management using python and flask as the backend and vuejs3 as the frontend.
It is a Show booking app which has multiple users and one Admin and Storemanagers. Users can buy products and can view them in their cart. The project also uses backend jobs using celery and the redis database to export jobs and do async jobs as well.

Technologies used

bcrypt==4.0.1
celery==5.3.6
Flask==3.0.0
Flask-Bcrypt==1.0.1
Flask-Caching==2.1.0
Flask-Cors== 4.0.0
Flask-Excel==0.0.7
Flask-Login==0.6.3
Flask-Principal==0.4.0
Flask-RESTful==0.3.10
Flask-Security-Too== 5.3.2
Flask-SQLAlchemy== 3.1.1
Flask-WTF==1.2.1
Jinja2==3.1.2
kombu==5.3.4
redis==5.0.1
requests==2.31.0
SQLAlchemy==2.0.23
urllib3==2.1.0
Werkzeug==3.0.1
WTForms==3.1.1

DB Schema Design

adminreq (id INTEGER NOT NULL, username VARCHAR(100), email VARCHAR(100), user_id INTEGER NOT NULL, is_approved BOOLEAN, PRIMARY KEY (id), UNIQUE (username), FOREIGN KEY(user_id) REFERENCES user (id))

cart (id INTEGER NOT NULL, name VARCHAR(100) NOT NULL, rate INTEGER NOT NULL, quantity INTEGER NOT NULL, amount INTEGER NOT NULL, item_id INTEGER NOT NULL, user_id INTEGER NOT NULL, is_sent BOOLEAN, PRIMARY KEY (id), FOREIGN KEY(item_id) REFERENCES item (id), FOREIGN KEY(user_id) REFERENCES user (id))

category (id INTEGER NOT NULL, name VARCHAR(100) NOT NULL, is_approved BOOLEAN, PRIMARY KEY (id))

item (id INTEGER NOT NULL, name VARCHAR(100) NOT NULL, stock INTEGER NOT NULL, sold INTEGER, manufactured_date DATETIME, date_added DATETIME, expiry_date DATETIME, rate INTEGER NOT NULL, category_id INTEGER NOT NULL, PRIMARY KEY (id), FOREIGN KEY(category_id) REFERENCES category (id))
role (id INTEGER NOT NULL, name VARCHAR(80), description VARCHAR(255), PRIMARY KEY (id), UNIQUE (name))

roles_users (user_id INTEGER, role_id INTEGER, FOREIGN KEY(user_id) REFERENCES user (id), FOREIGN KEY(role_id) REFERENCES role (id)) user (id INTEGER NOT NULL, username VARCHAR(100), email VARCHAR(100), password VARCHAR(100) NOT NULL, active BOOLEAN, fs_uniquifier VARCHAR(255) NOT NULL, PRIMARY KEY (id), UNIQUE (username), UNIQUE (fs_uniquifier)

API Design

The api for the app uses flask resource. Uses get methods to fetch items and categorys and even the user details. Uses the post method to add to the database. Put method to update existing entries and the delte method to delete entries.

Architecture and Features

To run the app first install and activate a virtual env and install all the python dependencies from requirement.txt in the /backend directory
After that run these commands in the /frontend directory
npm install -g @vue/cli
npm install bootstrap@5.3.2
npm install vue-router@4

then we can run the celery workers in the /backend directory
celery -A main:celery_app worker --loglevel INFO
celery -A main:celery_app beat --loglevel INFO

then we have to run the mailhog code to activate the port for sending emails
~/go/bin/MailHog

The app has one admin created upon running whose credentials are Admin@gmail.com pass- Admin123

The admin has to approve the user's request to become a storemanager before the user can use the manager panel

The admin has to approve any new categories created by the storemanager as well